

WEB APPLICATION DEVELOPMENT

DNA LAUNDRY



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Information System
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A.Introduction

DNA Laundry is a modern, customer-focused laundry service dedicated to delivering clean, neat, and timely garment care. Founded with the goal of simplifying everyday chores, DNA Laundry combines efficiency, hygiene, and technology to provide high-quality laundry solutions for individuals, families, and businesses.

With a commitment to Detail, Neatness, and Accuracy the very essence of our name DNA Laundry aims to set a new standard in the laundry industry. Whether it's everyday wear, delicate fabrics, or bulk orders, our team ensures every piece is handled with care and delivered fresh and spotless.

To support our growing customer base, DNA Laundry is equipped with a smart digital system that allows users to easily place orders, track their laundry status, and make payments online anytime, anywhere. This application simplifies the process of recording customer orders, laundry work status, and a more organized and efficient payment system. This application was created as part of the final project for the web-based programming course, with the aim of applying the concept of relationships between tables, as well as the CRUD (Create, Read, Update, Delete) process in real applications.

➤ Purpose of Application Development:

Simplify the process of inputting and managing laundry order data.

Provide transparency and tracking of laundry status to customers.

Improve the efficiency of laundry data management for business owners.

Apply the concept of database relations and business logic in PHP programming.

➤ Tools Used:

PHP: As the main backend programming language.

MySQL: For application data management and storage.

HTML&CSS: For the user interface.

Bootstrap For beautification and responsiveness.

XAMPP: As a local server to run the application.

B.Application Overview

➤ Main Function of Application

DNA Laundry application is a web-based system designed to facilitate customers in using laundry services. This application allows users to:

1. Register and record laundry orders
2. Monitor the status of the laundry process in real-time
3. Select and view details of service types and laundry packages
4. Confirm payment
5. View history and details of previous orders

This application supports CRUD (Create, Read, Update, Delete) features specifically for entities related to customer interaction such as orders, user accounts, and payment confirmations.

➤ User Type

This application is designed to be used by:

1. User / Customer
2. Register and login to the application
3. Create and manage laundry orders
4. View live laundry status
5. Confirming payment
6. View service history and delivery/collection status

➤ General App Workflow

1. Order / Pickup Registration

Customer creates a new order through the app

Select service type, package, and number of laundry items

2. Laundry Process

Laundry status will be updated automatically to “In Process”

Customers can monitor the status through the dashboard

3. Drying & Finishing

Once the washing process is complete, the status changes to “Completed”

4. Delivery / Pickup

Customers choose to pick up directly or wait for delivery

Order status is updated to “Picked Up” or “Delivered”

C. Table Information & Relationship

Table List

1. user

Stores customer data such as fullname, email, and password.

Additional columns: created_at, updated_at, status

Primary Key: id

foreign key: email

2. orders

Stores laundry order data created by the user.

Presumably has columns such as user_id, service_id, status, total, created_at

Foreign Key: user_id → refers to users.id

Foreign Key: service_id → refers to services.id

3. services

Stores data on the type of laundry service (for example: dry cleaning, ironing).

Primary Key: id

4. subscription_plans

Stores data on subscription plans that customers may use.

Primary Key: id

If the relationship with users is applied, then there may be a plan_id column in the users table.

5. remember_tokens

- Stores user authentication tokens (used for automatic login).

Usually related to users.id but not required to be described in the application's main relation.

Explanation of Inter-Table Relationships

users → orders

6. user_profil

foreign key: id

- One user can have many orders (one-to-many relation).

Relationship: orders.user_id → users.id

orders → services

One order is associated with one type of service.

Relation: orders.service_id → services.id

One user may have one subscription plan.

If there is a plan_id field in users, then the relation: users.plan_id → subscription_plans.id

users → remember_tokens

One user has one or more tokens for automatic login purposes.

Typical relation: remember_tokens.user_id → users.id (optional)

D.APPLICATION FLOW

1. Login / Authentication

User opens the application and accesses the login page.

User enters email and password.

The system performs verification:

If the data is valid, the user is redirected to the main dashboard.

If invalid, an error message is displayed.

2. Create Laundry Order

After login, the user accesses the “Create Order” feature.

The user fills in the order form:

Selects the type of service (refers to the services table)

Fills in the quantity or description of laundry items

(Optional) Select subscription package if available

The system saves the data to the orders table with an initial status, for example: Waiting for Processing.

3. Order Status Update

Admin or backend system updates the order status step by step:

- In Process: after the laundry is started.
- Completed: after the washing and drying process is complete.
- Delivered / Picked up: when the order is delivered to the customer or picked up directly.

Status updates are made to the status column in the orders table.

Every status change can be seen directly by the customer through the dashboard.

4. Completed & Confirmation

After the status changes to “Completed” or “Delivered”, customers can:

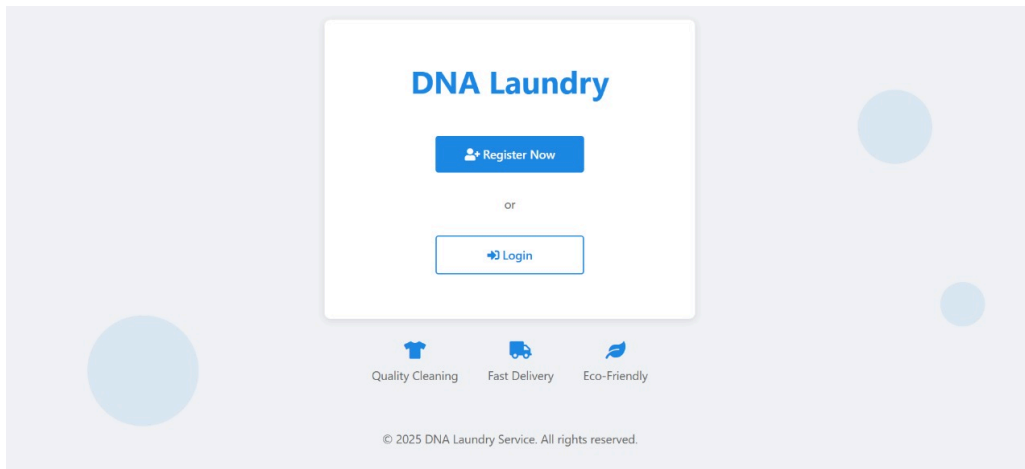
View the final order details

Confirm the payment if it has not been made yet

The system marks the order as completed and archives its history on the “Order History” pa

E.WEB VIEW

➤ Welcome page



The Welcome Page is an opening page that displays navigation buttons such as “Register” and “Login”. This page does not use CRUD functions because it only serves as an introduction to enter other parts of the system.

➤ Create Page

The Create Page or registration page allows users to create a new account. On this page there is a form to enter user data such as email and password. This page uses the Create function of CRUD because new user data is added to the system when the form is submitted

- Create

```
$sql_create_table = "CREATE TABLE IF NOT EXISTS users (  
  
    id INT(11) NOT NULL AUTO_INCREMENT PRIMARY KEY,  
  
    fullname VARCHAR(100) NOT NULL,
```

```

        email VARCHAR(100) NOT NULL UNIQUE,

        password VARCHAR(255) NOT NULL,

        role ENUM('user', 'admin') DEFAULT 'user',

        created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

        updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
    );

```

```

if ($conn->query($sql_create_table) === FALSE) {

    die("Error creating table: " . $conn->error);

}

```

```

$sql_create_profile_table = "CREATE TABLE IF NOT EXISTS user_profiles (

    id INT(11) NOT NULL AUTO_INCREMENT PRIMARY KEY,

    user_id INT(11) NOT NULL,

    phone VARCHAR(20),

    address TEXT,

    profile_image VARCHAR(255),

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,

    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE

);

```

```

if ($conn->query($sql_create_profile_table) === FALSE) {

    die("Error creating profile table: " . $conn->error);

}

```

```

$upload_dir = "uploads";

```

```

if (!file_exists($upload_dir)) {

    mkdir($upload_dir, 0777, true);

}

```

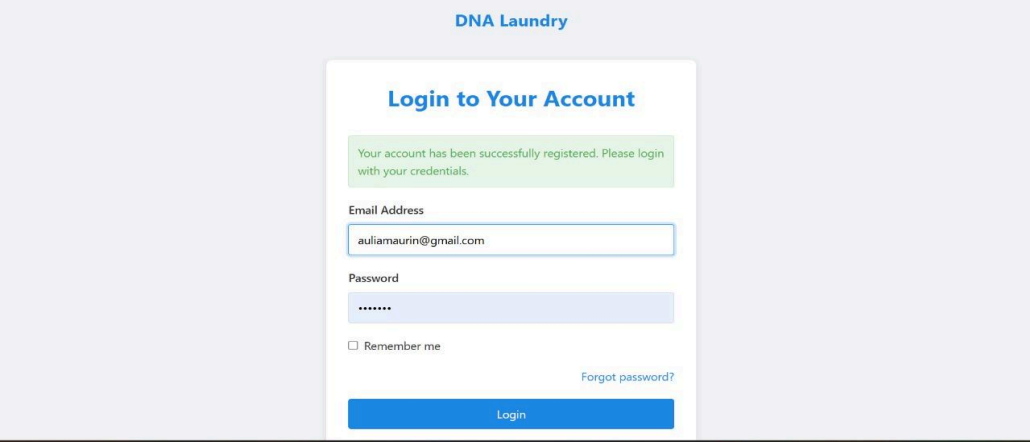
```
$profile_dir = "uploads/profiles";

if (!file_exists($profile_dir)) {

    mkdir($profile_dir, 0777, true);

}
```

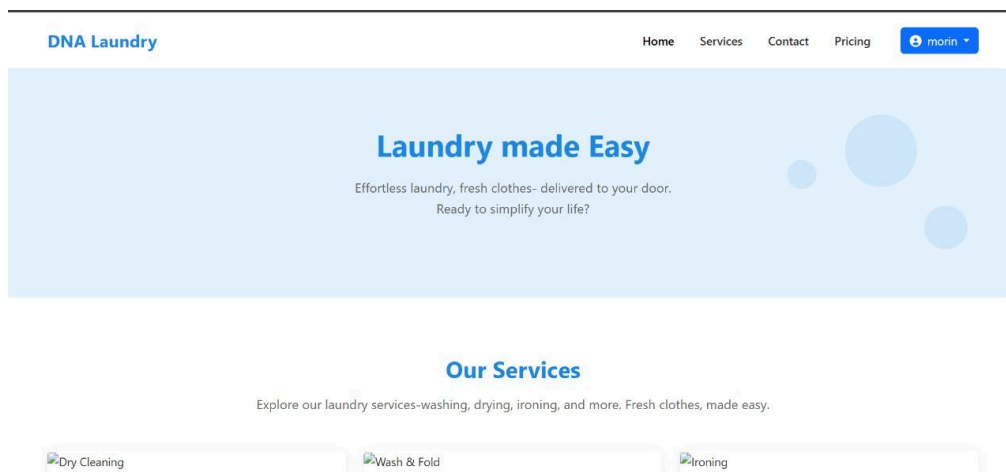
➤ Login Page



The screenshot shows the 'Login to Your Account' page for DNA Laundry. At the top, a green message box states: 'Your account has been successfully registered. Please login with your credentials.' Below this, there are two input fields: 'Email Address' with the value 'auiamaurin@gmail.com' and 'Password' with masked characters '*****'. A 'Remember me' checkbox is present below the password field. To the right of the password field is a link for 'Forgot password?'. At the bottom of the form is a blue 'Login' button.

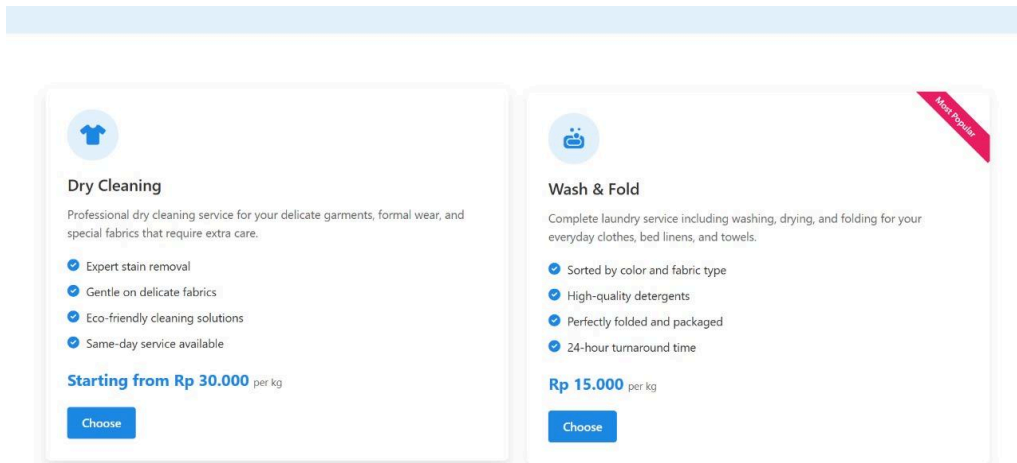
The Login Page is used for the user authentication process. Users must enter their email and password to log in to the system. Although important, this page is not included in CRUD operations as it does not add, read, update, or delete stored data directly.

➤ Home Page



The Home Page displays general information and services offered by DNA Laundry. There are navigation menus such as Home, Services, Contact, and Pricing. This page is only informative and does not involve any direct CRUD function

➤ Service Page



The Service Page displays a list of available laundry services, including service descriptions and prices. This page uses the Read function as it displays the service data from the system. If these services can be added, updated, or deleted by the admin through other pages, then the Create, Update, and Delete functions also apply, although they are not directly visible in this view.

➤ Orders Page

The screenshot displays the 'Orders Page' with a light blue header. Below the header, there is a 'Place Your Order' section with the text 'Fill in the details and we'll take care of the rest'. Below this, there are two main sections. The left section is a form for placing an order. It has a 'Select Service' dropdown menu with 'Dry Cleaning (Rp 30.000/kg)' selected. Below it is an 'Estimated Weight (kg)' input field with '1' entered. A note says 'Final weight will be confirmed after pickup'. There are two more input fields: 'Pickup Date' with 'hh/bb/yyyy' and a calendar icon, and 'Pickup Time' with 'Choose a time...' and a dropdown arrow. The right section is an 'Order Summary' box. It shows 'Service: Dry Cleaning', 'Price per kg: Rp 30.000', and 'Estimated Weight: 1 kg'. At the bottom, it shows 'Total: Rp 30.000' and a blue 'payment' button.

The Orders Page is where users can place an order for laundry services. Here the user selects the type of service, enters the estimated weight of the laundry, and specifies the pick-up date. After all the data is filled in, the user can press the payment button. This page uses the Create function because new order data is added to the system. If the user can change the data before payment, then the Update function can also apply.

- Update

```
if ($_SERVER['REQUEST_METHOD'] == 'POST') {

    // Dapatkan nilai original dari database

    $originalAddress = $order['address'];

    $originalServiceName = $order['service_name'];

    $originalWeight = $order['weight'];
```

- Delete

```
$deleteSql = "DELETE FROM orders WHERE order_id = ? AND user_id = ?";

$stmt = $conn->prepare($deleteSql);

$stmt->bind_param("ii", $orderId, $_SESSION['user_id']);

$stmt->execute();

$stmt->close();
```

➤ My Orders Page

DNA Laundry
Home
Services
Contact
Pricing

My Orders

Manage your laundry orders

#	Service	Weight	Pickup	Delivery	Total	Actions
ORD-A623C497	Dry Cleaning	1.00 kg	15 May 2025	17 May 2025	Rp 30.000	Edit Cancel

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The My Orders Page displays a list of user orders in tabular form. On this page, users can view order details such as service type, laundry weight, pick-up and delivery dates, and total cost. There are also action buttons such as “Cancel” to delete the order. Therefore, this page uses the Read function to display data, and Delete to delete certain orders.

- Select

```
$sql = "SELECT * FROM orders WHERE user_id = ? ORDER BY order_date
DESC";

$stmt = $conn->prepare($sql);
```

```
$stmt->bind_param("i", $userId);  
$stmt->execute();  
$result = $stmt->get_result();  
$orders = [];
```

F.Source Code

GROUP 9-Maurin,Nabila and Auliyah

G.Problem and Solution

1. Problem: Can't Connect to Database

When running the application, the page displays an error

Cause: Incorrect MySQL username/password.Database has not been created or connected.

Solution:Checking if the database `dna\_loundry` already exists in phpMyAdmin.

2. Problem:Data Not Logged into Database

Description:After submitting the form, the data is not saved.

Causes:Incorrect SQL query or empty variable,There is no `name=""` in the form input,Not handling the POST method correctly.

Solution: Double check the PHP script

H.Improvement

1. Smart Recommendation Feature

Using AI to provide personalized recommendations based on user preferences and usage history

2. Points & Reward System

Users earn points every time they contribute or use certain services, which can be exchanged for gifts or other benefits.

3. Multiplatform Integration

Availability of app versions for iOS, Android, and Web with real-time data synchronization.

4. Smart Notifications

Contextual notifications that only appear when they are truly relevant to the user's activity or interests.

5. Community Features & Discussion Forum

A place for users to discuss, ask questions, and share experiences or suggestions.

6. Activity Analytics and Reports

Personal usage statistics, trend insights, and user progress in the form of graph visualizations.

7. Offline Mode

Access some key features without an internet connection, with automatic synchronization when online.

8. Multilingual Support

To reach users from different backgrounds and regions.

If you can tell me more about the main functions of BERIKAN, I can suggest more specific and context-appropriate features.